

```
*| > SU_1_3
*| > u_sul_3

*| Version 1.30 created by Andy 1986.09.16

*| Change log

*| 1.30 1986.09.16 Kosher for second processors, network and ADFS
*|       except there appears to be a major bug somewhere
*| 1.31 1987.03.11 Two small bug fixes, major mods to file access.

*| Note : not kosher for communicator on these counts :
*| (1) needs to replace PROCasm with an 816-able approach
*| this is fairly easily solved - also see module in TEMP
*| (2) cannot BGET# or BPUT# from 64xx
*| this could be soluble if we can rely on resident integers being at
*| PAGE-some defined offset.
*| NB I cannot see a solution to either problem which is portable between
*|     BBC, Master, 2nd 6502 and Communicator.
*| the implication of the latter statement is that I was being particularly
*| dim at the time, since (2) is easily solved.

*| Changes @ 1.3 : Creation of PROCbget and PROCbput necessitate all I/O to
*|       random access files being done through indirection operators
*|       since PROCs cannot return values through their parameters
```

AUTO

*| (C) Pennine Software 1986, 1987

MODE7:A%=HIMEM:HIMEM-A%-63000

```
PROCsetup
PROCfiles
PROCfirst
```

REPEAT:PROCinline:UNTILend

```
PROCnodes
PROCTidyup
```

```
PROCcoords
END
```

```
DEFFROCSkip(K%):LOCAL C%,J%
*****
*| skip K% characters from input file OR point to <cr> if it occurs
*|     before K% characters have been read
*| Locals : C% used to receive characters from file
*|       J% byte count within the field
*|
*****
J%=0:REPEAT:C%=BGET#I%:J%=J%+1:UNTILC%=13ORJ%=K%
IFC%=13THEN PTR#I%=PTR#I%-1
ENDPROC
```

```
DEFFPROCread6(K%):LOCAL J%
*****
*| read6 : read a string of to six characters into $K% ( padding to right with
*|       spaces and leaving pointer pointing to <cr> if it occurs before six
*|       characters have been read )
*| Local : J% used as a byte count within the field
*|
*****
J%=1:REPEAT:J%=J%+1
K%?J%=BGET#I%:UNTIL(K%?J%)=13 ORJ%=5
IF(K%?J%)=13 THEN PTR#I%=PTR#I%-1
IFJ%<5 FORJ%=J%TO5:K%?J%=32:NEXT
ENDPROC
```

```
DEFFNreadn(K%,convert%):LOCAL N$,N,C%,J%
*****
*|
*| readn : read a number from a K% character field ( if <cr> occurs before K%
*|       characters have been read, leave pointer pointing at <cr> ) such
*|       that a blank or null-length field gives result zero
*| If convert% has bit 0 set then convert to deca-microns
*| If convert% has bit 1 set then convert from feet
*| If convert% has bit 2 set then return as a real else an integer
*|
*| Locals : N$ accumulates characters from the line, then used to be EVALled
*|       N holds the real result of the conversion
*|       C% used to receive BGETted characters from the file
*|       J% used as a byte count within the field
*|
*****
```

```
N$="" :C%=BGET#I%:IFC%=13THEN PTR#I%=PTR#I%-1:0
J%=1:REPEAT:N$=N$+CHR$(C%):C%=BGET#I%:J%=J%+1:UNTILC%=13ORJ%=K%
IFC%<>13N$=N$+CHR$(C%):ELSE PTR#I%=PTR#I%-1
IF N$=STRING$(LENN$, " ") THEN =0
N=EVAL(N$):IFconvert%AND1 N=100000*N
IFconvert%AND4 IFINFEET% N=N*.3048
IFconvert%AND2 THEN =N ELSE:=INT(N*0.5)
```

```
DEFFNrest:LOCAL G%
*****
*|
*| rest : return the rest of the line up to any control character as a string.
*|       note that if the pointer was pointing at <cr> this is a null string
*|       the pointer points at the start of the next line
*| Local : G% used to receive characters from the file
*| Features : No checking for excessively long lines, so could trap with
*|       "String too long"
*|       Terminated by any control character - desirable but non-standard
*|
*****
rest$="":G%=BGET#I%:IF G%<32 THEN =" "
REPEAT:rest$=rest$+CHR$(G%):G%=BGET#I%:UNTILG%<32
=rest$
```

```
DEFFNyn:LOCAL C%
*****
*|
*| FNyn : wait for user to press Y or N in either case. Return TRUE if it was Y
*| Local : C% used as a result register for the INString
*|
*****
REPEAT:C%=INSTR("yYnN","GET$):UNTILC%<0
=C%<3
```

```
DEFFPROCsetup
*****
*|
*| setup : define fixed constants, title, keywords, flags, offsets, and create
*|       machine code support.
*| Globals : RELEASE used to print identity info at various stages
*|       end remains FALSE until program is to finish
*|       KEYS% points to buffer holding valid keywords
*|       F%, T% point to buffers for survey station names ( from & to )
*|       STA% points to buffer for survey station table
*|       A% retains value of HIMEM from program start
*|       oldI$ string array to hold names of temporarily closed files
*|       oldP% integer array to hold their PTR# values
*|       UWATER% flag controls whether data are for dry or diving survey
*|       INFEET% flag controls whether data are to be converted
*|       DTHETA amount to be SUBTRACTED from bearings to get true North
*|       DPHI amount to be subtracted from clino reading to get truth
*|       D%, L% accumulative total and plan lengths of survey
*|       rest$, COMMENT$ - used throughout to get strings from file
*|
*****
```

```
@%-690A:RELEASE=1.30:end=FALSE
FORJ%=0TO1
PRINTCHR$141;CHR$131;"PENNINE SOFTWARE CAVE SURVEY ";RELEASE
NEXT:PRINT"
DIM KEYS% 56,F% 7,T% 7,W% 15,a% 15,b% 15
DIM oldI$(4),oldP$(4)
STA%=HIMEM:@%-690A
FORI%=HIMEM TO A%-4STEP4:I%=0:NEXT
FOR J%=0TO4:oldI$(J%)=" " :NEXT
$KEYS%="ORIGIN":$(KEYS%+7)="FEET " :$(KEYS%+14)="METRIC":$(KEYS%+21)="DIVING":$(K+
EYS%+28)="NORMAL":$(KEYS%+35)="CHANGE":$(KEYS%+42)="FILE " :$(KEYS%+49)="END "
F%*6-13:F%*77=0:T%*76=13:T%*77=0
UWATER%=FALSE:INFEET%=FALSE
DTHETA=0.:DPHI=0.:D%=0:L%=0
rest$=STRING$(36, " ") :COMMENT$=rest$
PROCasm
VDU28,0,23,39,2
ENDPROC
```

```
DEFFPROCfiles
*****
*|
*| files : get filenames from user - check they exist ( where possible )
*|
*| Globals : d$ holds path name to reach SU subdirectories.
*|       note that this is restricted by the filing system in use, and
*|       may take various values, including ( for masters ) fs name.
*|       I$ primary input filename
*|       level% current nesting level of subsidiary input files
*|       O$ output filename - defaulted same as input.
*|       I%, O% handles of open input and output files
*|
*****
ON ERROR OFF:REM error trapping corrupts stack anyway. need M/C.
REPEAT
```

```
INPUT"Path name : "d$:IFd$<>""d$=d$+ "."
PRINT"Primary input file ";d$;:INPUT"D."I$
I%=OPENIN(d$+"D."+I$):IF I%=0 PRINT""Unable to find this file""Try again.""
UNTILI%<0
level%=0
INPUT "Output filename "O$
IFO$=""O$=I$
O%=OPENOUT(d$+"O."+O$)
IFO%=0 P.'.' "Unable to open ";d$;"O.";O$' "possibly the directory does not exist." :+
END
VDU12,26
FORJ%=0TO1:PRINTCHR$141;CHR$130:NEXT
VDU28,0,23,39,2
ENDPROC
```

```
DEFFPROCfirst:LOCAL G%
*****
*|
*| first : read set-up from input file - title, origin coordinates
*|       Globals : title$ holds a title for the current survey, printed at times
*|       STA%$0 contains the name of the primary origin station
*|       S% a running count of the survey station number from zero
*|       a%!... x,y,z metric coordinates of origin ( internal format )
*|
*| Change @ 1.3 : a%! used instead of A%.B%.C%
*|
*****
```

```
title$=FNrest
PRINT"title$' STRING$(LENTtitle$,"=)
PROCread6(STA%)
STA%*76=13
STA%*77=0:REM No. of references
S%=0:REM station count
PROCSkip(6)
Ia%=FNreadn(12,1)
a%*14=FNreadn(12,1)
a%*18=FNreadn(12,1)
COMMENT$=FNrest
PRINTCHR$130"ORIGIN at "CHR$134;$STA%
IFLENCOMMENT$PRINTCHR$134COMMENT$
ENDPROC
```

```
DEFFPROCasm
*****
*|
*| asm : assemble machine code support for station name search
*| Globals : code% points to DIMensioned space for the machine code
*|
*| NB all internal labels may as well be local.
*| NB portability requirements mean we will have to eliminate the use of fixed
*|     location resident integer variables. This shouldn't be a problem.
*|
*| BUG fix 1987.004 end of search was determined by compare with zp 684
*|     which was supposed to be set to HIMEM. It has never been initialised,
*|     and why I don't just do CMP #(HIMEM DIV 256) ?? Well, I do now.
*|
*****
```

```
DIM code% 127
FOR J% = 0 TO 2 STEP 2
P%=code%
[OPTJ%
LDA 642C \ K% LSB points to station to search for
STA 682
LDA 642D
STA 683
LDA 6438 \ N% LSB points to last station in table
STA 680
LDA 6439
CLC
ADC #(HIMEM DIV 256)
STA 681
.nextsta%
LDY #5
.nextbyte%
LDA 680,Y
CMP 682,Y
BNE notthisone%
DEY
BPL nextbyte%
LDA 681
SEC
SBC #(HIMEM DIV 256)
STA 642D
LDA 680
STA 642C
RTS
.notthisone%
LDA 680
BNE cantbeEND%
LDA 681
```

<unset> <unset>

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```
CMP #(HIMEM DIV 256)
BNE notEND%
LDA #6FF
STA #42F
RTS
.notEND%
DEC #81
LDA #80
.cantbeEND%
SEC
SBC #8
STA #80
JMP nextsta%
]
NEXT
ENDPROC

DEFFPROCinline:LOCAL K%
*****
*| inline : process a line from the current input file
*| Globals : F%, T%      point to buffers for new station names
*|          KEYS%       buffer for keywords to check against "from" station
*|          K%          index of keyword, if matched. -1 => unmatched
*|
*****
IFEOF#I%PROCend:ENDPROC
PROCread6(F%)
K%=-1:FORJ%=-0TO49STEP7
IFSF%=$(KEYS%+J%) K%=J%/7
NEXT
IFK%<0 PROCdata:ENDPROC
IFK%=0 PROCorigin:ENDPROC
IFK%<3 PROCunits:ENDPROC
IFK%<5 PROCstyle:ENDPROC
IFK%=5 PROCchange:ENDPROC
IFK%=6 PROCnfile:ENDPROC
IFK%=7 PROCend:ENDPROC

DEFFPROCend
*****
*| end : process the END keyword in the input file
*| Globals : I%          handle of currently open input file
*|          level%       current depth of nesting - if zero then this is the
*|                      primary input file that we have open.
*|          end          global flag to stop 1st phase of SU on closing primary
*|          K%           keyword index. will be 7 if we are here because of an
*|                      END keyword, will be some other value if the file just
*|                      ended on us.
*| BUG fix 1987.001      K% might have a value > -1 if the last line contained
*|                      a keyword. Condition changed to K%=7, from K%>0
*|
*****
CLOSE#I%
IF level%=0 end=TRUE:ENDPROC
IF K%=7 PRINT""END occurs in nested file "I$""Halt processing at this level ?";+
IF FNyn end=TRUE:ENDPROC
level%=level%-1
I$=oldI$(level%)
I%=OPENIN(d$+"D."+I$)
PTR#I%=oldP$(level%)
ENDPROC

DEFFPROCnfile
*****
*| nfile : save place in old file and open nested file for input
*| Globals : level%       current nesting level. limited arbitrarily to 4
*|          oldI$, I$     old and new filenames
*|          oldP%         PTR# of old file, so we can resume input later
*|
*****
IFlevel%>4 PROCthrowwobblor(1):ENDPROC
oldI$(level%)=I$:I$=FNrest
oldP$(level%)=PTR#I$:CLOSE#I%
I%=OPENIN(d$+"D."+I$)
IFI%=0 PRINT"Unable to open new input file""d$+"D.";I$:I$=oldI$(level%);I%=OPENIN(
(d$+"D."+I$):PTR#I%=oldP$(level%);PRINT"Input continues from "d$;D.";I$:ENDPROC
level%=level%+1:ENDPROC

DEFFPROCorigin
*****
*| origin : read data defining alternate origin
*| Globals : T%          station name for new origin is read into "to" buffer
*|          X%, Y%, Z%   x,y,z coordinates of new origin (internal format)
*|          U%           index of new station name in existing list ( should be
*|                      -1, if not, user is redefining an existing station )
*|          S%           internal number for new survey station
*|
*****
STA%$U%      new name added to list
STA%?(U%+7)  count of references to new station - initially zero
W%           32 bit value with high word set to -1 and low word set
             to station no.: S%. Forms from/to for origin definition
*| Note : W% is set to U%/8-65536, not S%-65536, since "origin already defined"
             is treated as a soft error, so U%=S%*8 not always true !
*|
*| Bug fix 1987.003      N% assigned to valid value before calling FNscan.
*|                      I cannot see how the program ever worked before.
*|
*| Change @ 1.31 : use of PROCbput. eliminates W% in this context.
*|
*****
PROCskip(30):COMMENT$=FNrest
PROCread6(T%):PROCskip(6)
X%=FNreadno(12,5)
Y%=FNreadno(12,5)
Z%=FNreadno(12,5)
N%=8*S%
U%=FNscan(T%)
IF U%<0 S%=S%+1:U%=S%*8:$ (STA%+U%)=$T%:STA%?(U%+7)=0 ELSE PROCthrowwobblor(2)
*|W%=U%/8-65536
PROCbput(0%,4,U%/8-65536,X%,Y%,Z%)
*|FORJ%=445C TO446B:BPOT#0%,?J%:NEXT
IFLENCOMMENT$PRINT'CHR$132COMMENT$
COMMENT$=FNrest
*| we don't print this comment - would it be worth doing so ?
ENDPROC

DEFFPROCstyle
*****
*| style : process keywords to change type of input data
*| Globals : K%          index to keyword - must be 3 for DIVING, 4 for NORMAL
*|          UWATER%      flag to indicate current style
*|
*****
UWATER%=(K%=3)
IFUWATER% PROCdive:PROCskip(12) ELSE PROCskip(30)
COMMENT$=FNrest:IFLENCOMMENT$PRINT'CHR$129COMMENT$
ENDPROC

DEFFPROCdive
*****
*| dive : start surveying underwater type of data - get starting info
*| Globals : T%          points to buffer for starting station name
*|          Q%           internal station index for starting station
*|          depth         depth of most recently defined diving survey station
*|
*| Bug fix 1987.003      N% assigned before calling FNscan
*|
*****
PROCread6(T%)
N%=8*S%
Q%=FNscan(T%)
IFQ%<0 PROCthrowwobblor(3)
depth=FNreadno(6,7)
ENDPROC

DEFFPROCchange
*****
*| change : accept data to define new instrument calibrations
*| Globals : DTHETA       amount to be subtracted from compass to get true north
*|          DPHI          amount to be subtracted from clino reading to get truth
*|
*****
PROCskip(12)
DTHETA=FNreadno(6,2)
DPHI=FNreadno(6,2)
PROCskip(6)
COMMENT$=FNrest
IFLENCOMMENT$PRINT'CHR$134COMMENT$
ENDPROC

DEFFPROCthrowwobblor(E%):LOCAL@%
*****
*| throwwobblor : handle errors
*| Globals : T%          points to buffer holding erroneous station name
*| Locals : E%           error number
*|          @%           why is this LOCAL ?
*|
*****
PRINT'STRINGS(80,"*"):VDU7,7
PRINT" Error on input"
IF E%=1 PRINT"Input file nesting level too great""( maximum depth 4 )"
IF E%=2 PRINT"New origin redefines existing station : ";$T%
```

```
IF E%=3 PRINT"Depth change > leg length in dive"
IF E%=4 PRINT"Diving survey start station ";$T%," undefined or missing"
PRINT'STRINGS(80,"*"):VDU7
IF E%<4 STOP ELSE ENDPROC
```

```
DEFFPROCdata
*****
*| data : read and convert a survey leg - of either style
*| Globals : F%, T%      pointers to buffers for from and to station names
*|          G%, U%       pointers to from and to station names in main list
*|          L, THETA, D   values of true length, true bearing and plan length.
*|          X%, Y%, Z%   x,y,z vector of current survey leg
*|          D%, L%        cumulative plan and total length of survey
*|
*| Change @ 1.3 : Use of PROCbput. eliminates X%, Y%, W% in this context
*|
*****
PROCread6(T%)
PROCLIST(F%,T%)
IFPOS>34PRINT
IFU%>G% PRINT;$T%," "; ELSE PRINT;$F%," ";
L=FNreadno(6,7)
THETA=FNreadno(6,2)-DTHETA
IF UWATER% PROCdiving ELSE PROCdry
*|X%=INT(D*SINRAD(THETA)+0.5)
*|Y%=INT(D*COSRAD(THETA)+0.5)
D%=D%+INT(D+0.5)
L%=L%+INT(L+0.5)
COMMENT$=FNrest
*|FOR J%=445C TO 446B
*|BPOT#0%,?J%:NEXT
PROCbput(0%,4,8192*G%+U%DIV8,INT(D*SINRAD(THETA)+0.5),INT(D*COSRAD(THETA)+0.5),Z%)
)
IFLENCOMMENT$PRINT'CHR$131COMMENT$
ENDPROC
```

```
DEFFPROCdry
*****
*| dry : handle data for above water surveying style
*| Globals : PHI, PLUMB values of corrected clino reading and vertical plumb
*|          L, D, Z%      true length, plan length and vertical component of leg
*|
*****
PHI=FNreadno(6,2)-DPHI
PLUMB=FNreadno(6,7)
D=L*COSRAD(PHI)
Z%=INT(L*SINRAD(PHI)+PLUMB+0.5)
ENDPROC
```

```
DEFFPROCdiving
*****
*| diving : handle data for underwater surveying style
*| Globals : depthF, depthT      depth of from and to stations
*|          Q%, G, U%            pointers to station names for 'last new', 'from' & 'to'
*|          Z, L                 depth change and true length of survey leg
*|
*****
depthF=FNreadno(6,7)
depthT=FNreadno(6,7)
IFQ%=G%AND depthF=0 depthF=depth ELSEIFQ%=U%AND depthT=0 depthT=depth
IFQ%=G% depth=depthT:Q%=U% ELSE IFQ%=U% depth=depthF:Q%=G% ELSE IF G%>U% depth=de+
pthF:Q%=G% ELSE depth=depthT:Q%=U%
Z=depthF-depthT
IFABSZ>L PROCthrowwobblor(3)
D=SQR(L*L-Z*Z)
Z%=INT(Z+0.5)
ENDPROC
```

```
DEFFPROCLIST(F%,T%):LOCAL N%:N%=8*S%
*****
*| list : check both survey stations for existence in our list. Add them if
*| they aren't already there, and return internal station numbers
*| Globals : F%, T%      pointers to buffers holding 'from' and 'to' names
*|          G%, U%       offsets to location of these names in the internal list
*|          S%           number of current highest station
*|          Local : N%    offset to next free space in internal station list
*|
*| BUG fix 1987.002      If both stations are undefined, they both got added to
*| the list in the same place - ie. 'from' station failed to be added.
*| fixed by incrementing N% if from station has to be added.
*|
*****
```

```
G%=FNscan(F%)
U%=FNscan(T%)
*|IF G%<0 S%=S%+1:G%=N%+8:$ (STA%+G%)=$F%:STA%?(G%+7)=1 ELSE STA%?(G%+7)=STA%+(G%+
7)+1
```

```

IF G%<0 S%=S%+1:G%=N%+8:N%=G%:$ (STA%+G%)=SF%:STA%? (G%+7)=1 ELSE STA%? (G%+7)=STA%?+
(G%+7)+1
IF U%<0 S%=S%+1:U%=N%+8:$ (STA%+U%)=ST%:STA%? (U%+7)=1 ELSE STA%? (U%+7)=STA%? (U%+7)+
+1
*|W%=8192*G%+U%/8
ENDPROC

DEFFNscan(K%)
*****
*| scan : call machine code support to search for a station name
*| Note : uses knowledge of absolute location of resident integers - UGH !
*|
*****
CALL code%
=K%

DEFFPROCunits
*****
*| units : interpret FEET or METRIC keywords
*| Globals : K%      index to keyword used: 1 => INFEET, 2 => METRIC
*|          INFEET%  flag - true if we are converting from feet
*|
*****
INFEET%= (K%=1):PROCSkip(30)
COMMENT$=FNrest:IFLENCOMMENT$PRINT'CHR$129COMMENT$
ENDPROC

DEFFPROCnodes
*****
*| nodes : find all the nodes in the survey
*| Globals : d$, O$      pathname for all files, Output filename.
*|          O%           handle for output to node file
*|          J%, I%       count of stations, bytes
*|          J%?7        number of references to this station
*|          R%           offset to start of main station table in the file
*|          S%           station number of final station
*| on exit  R%           total number of nodes found
*|
*| Change @ 1.3 : Use of PROCbput.
*|
*****
CLOSE#O%:O%=OPENOUT(d$+"N."+O$)
FORI%=0TO7:BPUT#O%,0:NEXT
FORI%=0TO7:BPUT#O%,STA%?I%:NEXT
FORJ%=STA%+8TOSTA%+8*S%STEP8
IFJ%?7<>2 FOR I%=0TO7:BPUT#O%,J%?I%:NEXT
NEXT
R%=PTR#O%
FORJ%=0TO8*S%+7
BPUT#O%,STA%?J%
NEXT
PTR#O%=0
*|FORJ%=4444 TO444B:BPUT#O%,?J%:NEXT
PROCbput (O%,2,R%,S%,0,0)
CLOSE#O%
R%=R%/8-1
ENDPROC

DEFFPROCtidyup
*****
*| tidyup : close files, print statistics and get user to input setup info for
*|          final stage of survey
*| Globals : S%, R%, D%, L% - see caption printed with each one
*|
*****
@%=490A:CLOSE#0:REM just to make sure
VDU12,26
FORJ%=0TO1:PRINTCHR$141;CHR$134:NEXT
VDU28,0,23,39,2
PRINT'CHR$134;"STATISTICS : "
PRINT' "No. of survey stations ";S%+1
PRINT' "No. of nodes in survey ";R%
@%=42030A
PRINT' "Total plan length      "D%/100000;" m"
PRINT' "Total survey length  "L%/100000;" m""
SOUND1,-7,170,20
PRINT' DO YOU WANT OUTPUT PAGED ? ";IF F%yn VDU14 ELSE VDU15
PRINT' "" DO YOU WANT OUTPUT PRINTED ? ";IF F%yn VDU2 ELSE VDU3
CLS
PRINT' ",LEFT$(title$,38)'" ";STRING$(LENLEFT$(title$,38),"=")
IFLEN(title$>38 PRINT' ";MID$(title$,39)'" ";STRING$( (LEN(title$)-38,"=")
ENDPROC

DEFFPROCcoords
*****

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*|
*| coords : calculate and display coordinates and closure errors
*| Globals : d$, O$      path name for SU files, output filename to use
*|          O%, S%       handle for C. file, number of stations to reserve space
*|          STA%         base of internal station table
*|          A%, B%, C%   x,y,z coordinates of primary origin of survey
*|          I%           handle for input from previous output ( vector ) file
*|
*| Change @ 1.3 : Use of PROCbput
*|
*****
O%=OPENOUT(d$+"C."+O$):PTR#O%=12*S%-1:BPUT#O%,0:PTR#O%=0
STA%=HINEM
STA%?7=1:FORJ%=15TO12287STEP8:STA%?J%=0:NEXT
OSCLI("SPOOL "+d$+"P."+O$):@%=420309
*|FORJ%=4404TO440F:BPUT#O%,?J%:NEXT:L%=0
PROCbput (O%,3,A%,B%,C%,0)
I%=OPENIN(d$+"O."+O$)
PRINT' "Station      East/m North/m Height/m""
"
PRINT$STA%;TAB(6),A%/100000,B%/100000,C%/100000
REPEAT:PROCvector:UNTILEOF#I%
PROCcotidy
CLOSE#I%:CLOSE#O%:*SPOOL
ENDPROC

DEFFPROCvector
*****
*|
*| vector : calculate a new coordinate or closure from vector in file, or warn
*|          user that his stations are undefined
*| Globals : W%           pointer to 16 byte buffer.
*|          word 0 : hiword is 'from', lo is 'to' station no.
*|          word 1-3 x,y,z vector for this survey leg
*|          F%, T%      internal station numbers derived from W% in vector file
*|          C%, U%      offsets to station names in table
*|          G%?7, U%?7 specify whether each station is defined yet
*|
*| Change @ 1.3 : Use of PROCbget
*|
*****
*|FOR J%=445C TO446B:J%=BGET#I%:NEXT
PROCbget (I%,4,W%)
*|F%=(W%DIV65536:T%=(W%AND65535
F%=(I%W%)DIV65536:T%=(I%W%)AND65535
G%=STA%+8*F%:U%=STA%+8*T%
IFF%<0 PROCneworg:ENDPROC
K%=G%?7+2*U%?7
IF K%=1 PROCnewxyz (F%,T%, 1):PROCprint (U%):ENDPROC
IF K%=2 PROCnewxyz (T%,F%,-1):PROCprint (G%):ENDPROC
IF K%=3 PROCclosed (F%,T%):ENDPROC
PRINT$G%;" TO ";SU%:" is undefined."
ENDPROC

DEFFPROCnewxyz (D%,E%,V%)
*****
*|
*| newxyz : calculate new coordinate
*| Globals : L%           internal station number of last defined ( retained )
*|          W%           points to buffer containing x,y,z vector
*|          a%           points to buffer containing coordinates
*| Params : D%, E%       station numbers : defined & undefined
*|          V%           direction : +1 for fore (ie. D -> E ), -1 for backsight
*|
*| Change @ 1.3 : Use of PROCbput and PROCbget - with indirections
*|
*****
*|IF D%<>L% THEN PTR# (O%)=12*D%:FORJ%=4404 TO440F:J%=BGET#O%:NEXT
IF D%<>L% THEN PTR# (O%)=12*D%:PROCbget (O%,3,a%)
*|A%=A%+V%*X%:B%=B%+V%*Y%:C%=C%+V%*Z%
!a%=!a%+V%*W%!4:a%!4=a%!4+V%*W%!8:a%!8=a%!8+V%*W%!12
*|L%=E%:PTR#O%=12*E%:FORJ%=4404 TO440F:BPUT#O%,?J%:NEXT
L%=E%:PTR#O%=12*E%:PROCbput (O%,3,!a%,a%!4,a%!8,0)
ENDPROC

DEFFPROCneworg
*****
*|
*| neworg : define coordinates of alternate origin
*| Globals : W%           points to buffer with x,y,z - in this case coordinates
*|          a%           points to buffer containing the coordinates
*|          L%           station number of most recently defined station
*|          T%           station number of station 'to' ( the new origin )
*|
*****
*|A%=X%:B%=Y%:C%=Z%:L%=T%
!a%=W%!4:a%!4=W%!8:a%!8=W%!12:L%=T%
*|PTR#O%=12*T%:FORJ%=4404 TO440F:BPUT#O%,?J%:NEXT
PROCbput (O%,3,!a%,a%!4,a%!8,0)
PRINT'New origin ":"PROCprint (U%)

```

```

ENDPROC

DEFFPROCprint(G%)
*****
*|
*| print : print a set of coordinates
*| Global : a%           points to buffer containing coordinates to print
*| Parm   : G%           points to buffer containing station name and status
*|
*****
G%?7=1
PRINT$G%;TAB(6),!a%/100000,a%!4/100000,a%!8/100000
ENDPROC

DEFFPROCclosed(D%,E%)
*****
*|
*| closed : calculate and print a misclosure
*| Globals : L%           number of most recently defined station ( retained )
*|          a%           points to buffer containing coordinates or error
*|          W%           points to buffer containing vector of survey leg
*|          A%, B%, C%   new version of coordinates
*|          U%           points to buffer containing name of 'to' station
*| Note, though we pass the station numbers in D%,E% use of U% implies an
*| assumption that these are F%, T% respectively. This wants sorting.
*|
*****
*|IF D%<>L% THEN PTR# (O%)=12*D%:FORJ%=4404 TO440F:J%=BGET#O%:NEXT
IFD%<>L% PTR# (O%)=12*D%:PROCbget (O%,3,a%)
*|A%=A%+X%:B%=B%+Y%:C%=C%+Z%
A%=!a%+W%!4:B%=a%!4+W%!8:C%=a%!8+W%!12
*|PTR#O%=12*E%:FORJ%=4460 TO446B:J%=BGET#O%:NEXT
PROCbget (O%,3,W%+4)
PRINT$U%;TAB(6),A%/100000,B%/100000,C%/100000
*|A%=A%-X%:B%=B%-Y%:C%=C%-Z%
!a%=A%-W%!4:a%!4=B%-W%!8:a%!8=C%-W%!12
*|PRINT'previous:"X%/100000,Y%/100000,Z%/100000
PRINT'previous:"W%!4/100000,W%!8/100000,W%!12/100000
*|PRINT'MISCLOSES"A%/100000,B%/100000,C%/100000
PRINT'MISCLOSES"!a%/100000,a%!4/100000,a%!8/100000
L%=1:ENDPROC

DEFFPROCcotidy:*|LOCAL K%
*****
*|
*| cotidy : write out 'undefined' data for any undefined stations
*| Globals : a%           points to buffer for coordinates
*|          S%           number of final station in list
*| Note : the x,y,z 0,0,-maximum value is used to represent 'undefined'. Any
*| program which uses the coordinate data should ignore such stations
*|
*****
@%=490A:ENDPROC
*|A%=0:B%=0:C%=480000000
!a%=0:a%!4=0:a%!8=480000000
FORJ%=0TOS%
*|IF (8*J%?7)=0 THEN PTR# (O%)=(12*J%):FORK%=4404 TO440F:BPUT#O%,?J%:NEXT
IF (8*J%?7)=0 THEN PTR# (O%)=(12*J%):PROCbput (O%,3,!a%,a%!4,a%!8,0)
NEXT
ENDPROC

DEFFPROCbput (O%,N%,A%,B%,C%,D%):LOCAL J%
*****
*|
*| bput : put numbers into a file in minimum number of bytes, without the
*| built in inefficiency of PRINT#. There is still a built in problem
*| here which we could solve in version 3 or later BASIC, by having
*| !b%,b%!4,b%!8,b%!12 as the formal parameters.
*|
*****
!b%=A%:b%!4=B%:b%!8=C%:b%!12=D%
FOR J%=0 TO 4*N%-1
BPUT#O%,b%?J%
NEXT
ENDPROC

DEFFPROCbget (I%,N%,a%)
*****
*|
*| bget : reverse the effect of bput. Again, less efficient because we can't
*| return values in formal parameters.
*|
*****
FORJ%=0 TO 4*N%-1
a%?J%=BGET#I%
NEXT
ENDPROC

```